SVM & KNN Assignment

Day 33 | AM Session — Week 6, Machine Learning & AI

IIT Gandhinagar | PG Diploma in AI-ML & Agentic AI Engineering

Part A: Digit Samples, GridSearchCV Heatmap, Optimal K Selection, Confusion Matrices, Per-Class F1 Scores

Part B: FAISS vs sklearn Speed & Accuracy

Part C: Debug: Missing Scaler

Part D: SVM Decision Boundary (C sweep)

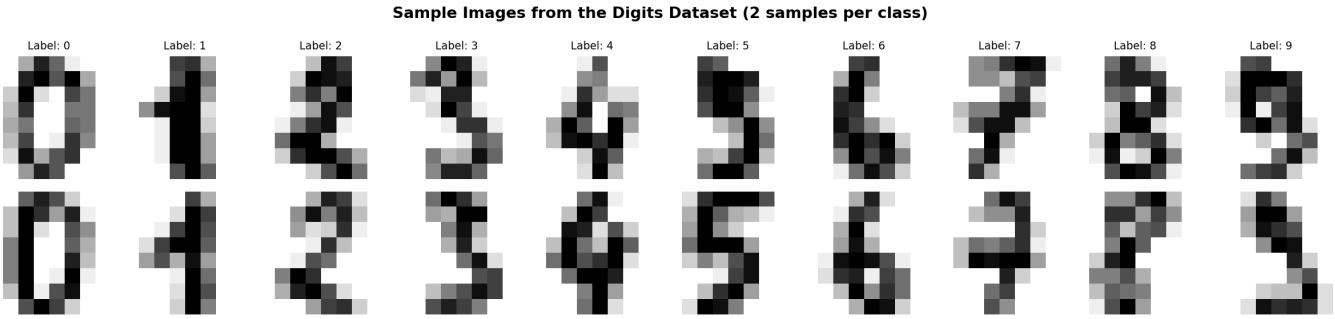

Figure 1: Sample Images from the Digits Dataset

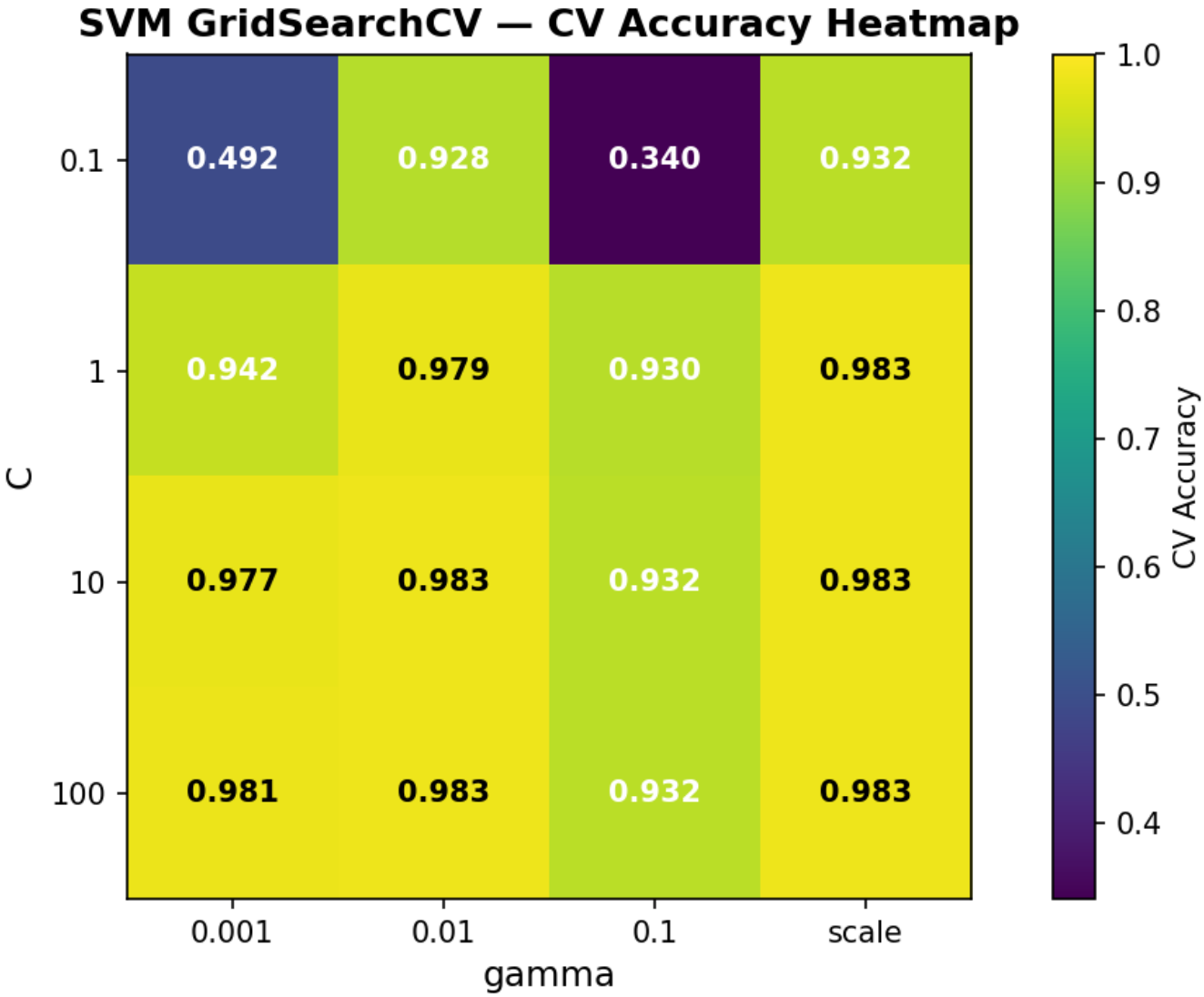

Figure 2: SVM GridSearchCV — CV Accuracy Heatmap (C × gamma)

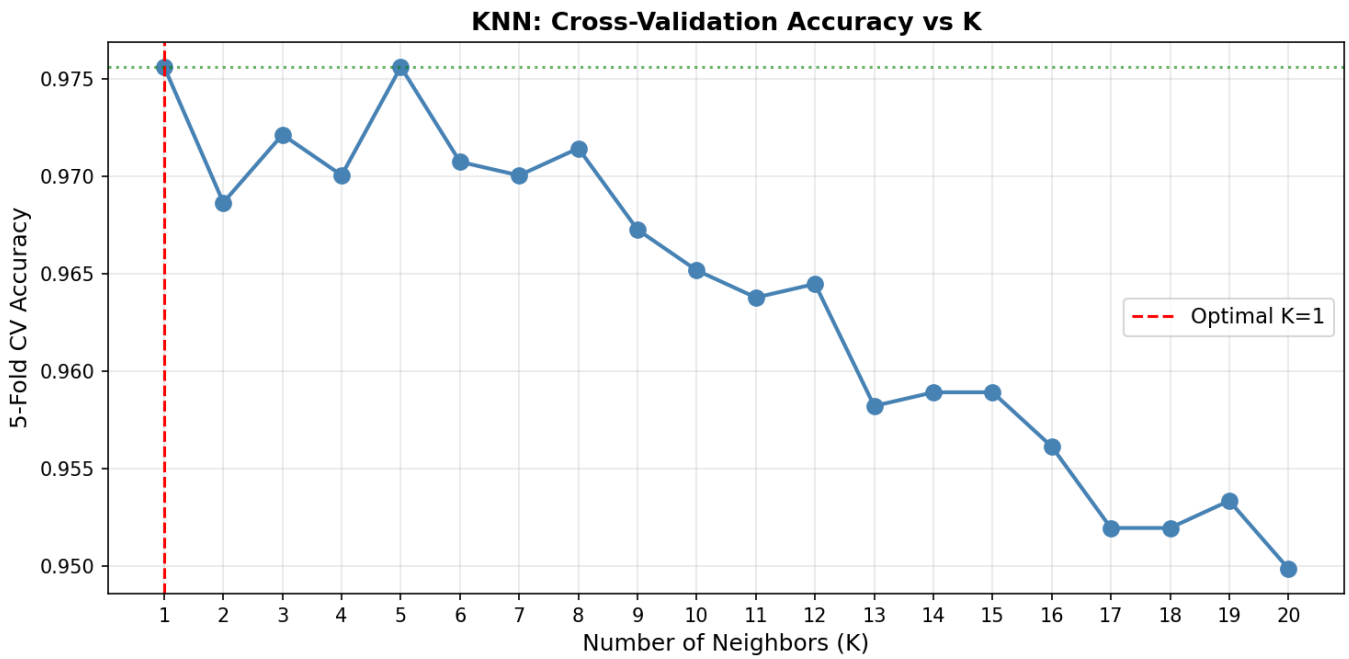

Figure 3: KNN — Cross-Validation Accuracy vs K

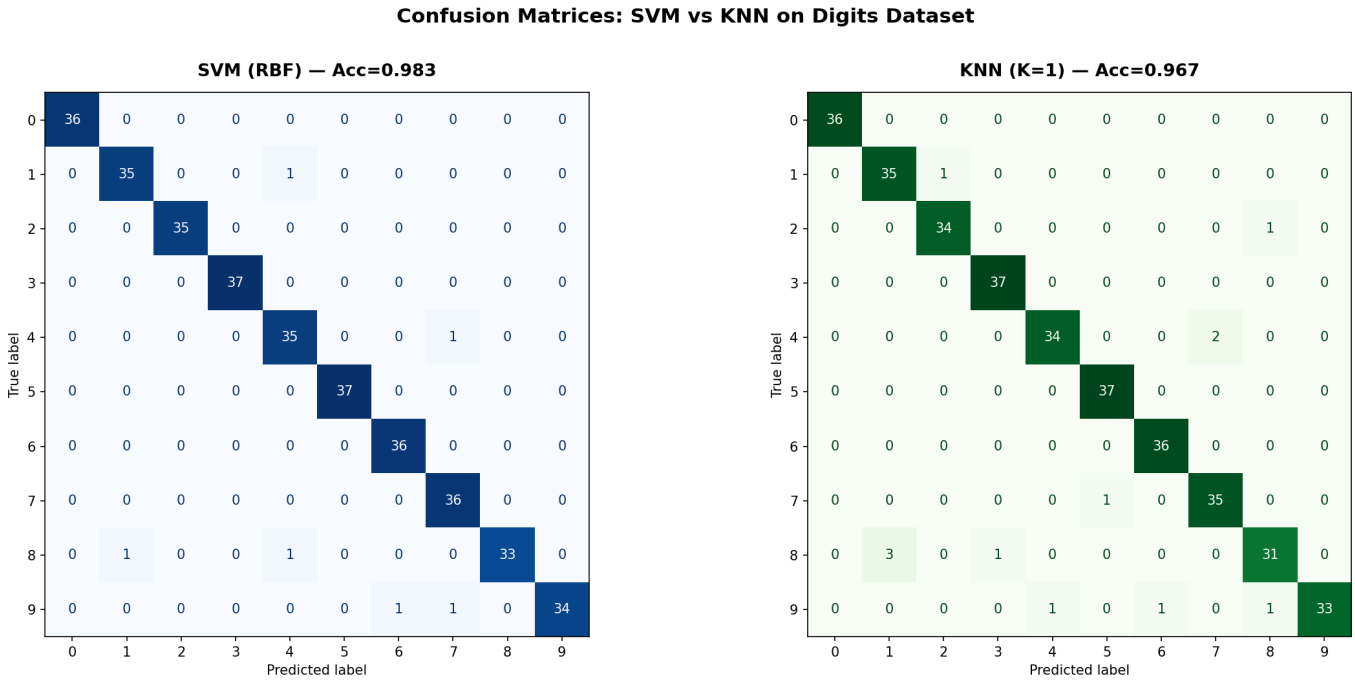

Figure 4: Confusion Matrices — SVM vs KNN

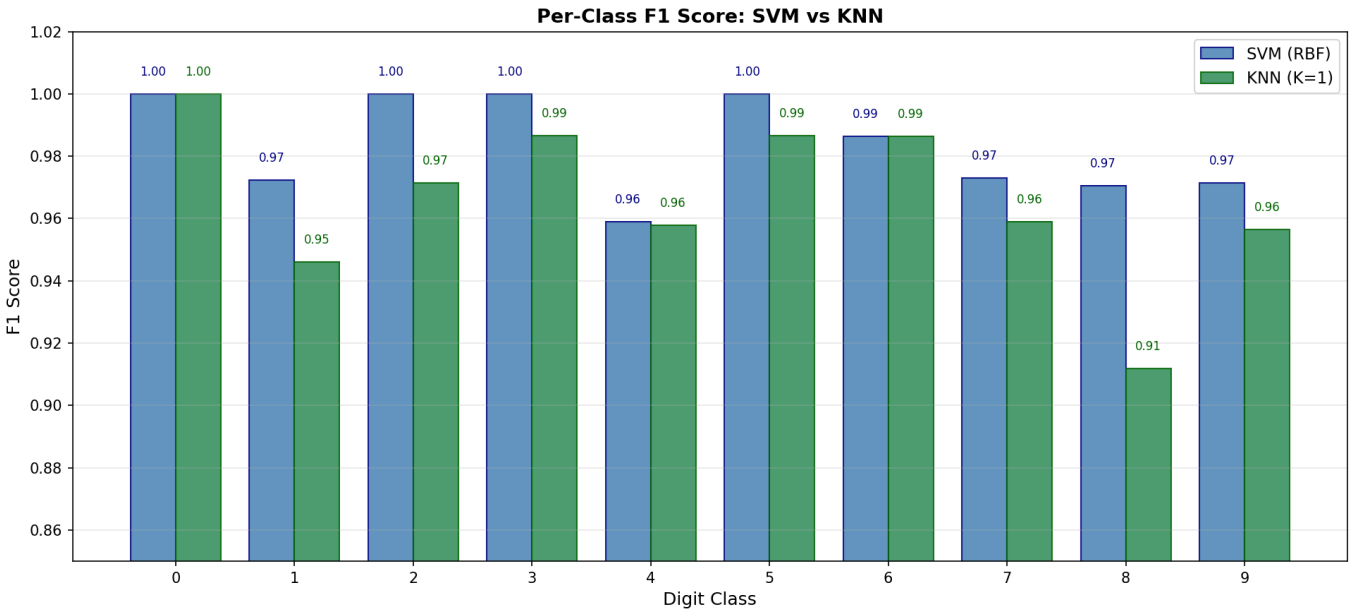

Figure 5: Per-Class F1 Score — SVM vs KNN

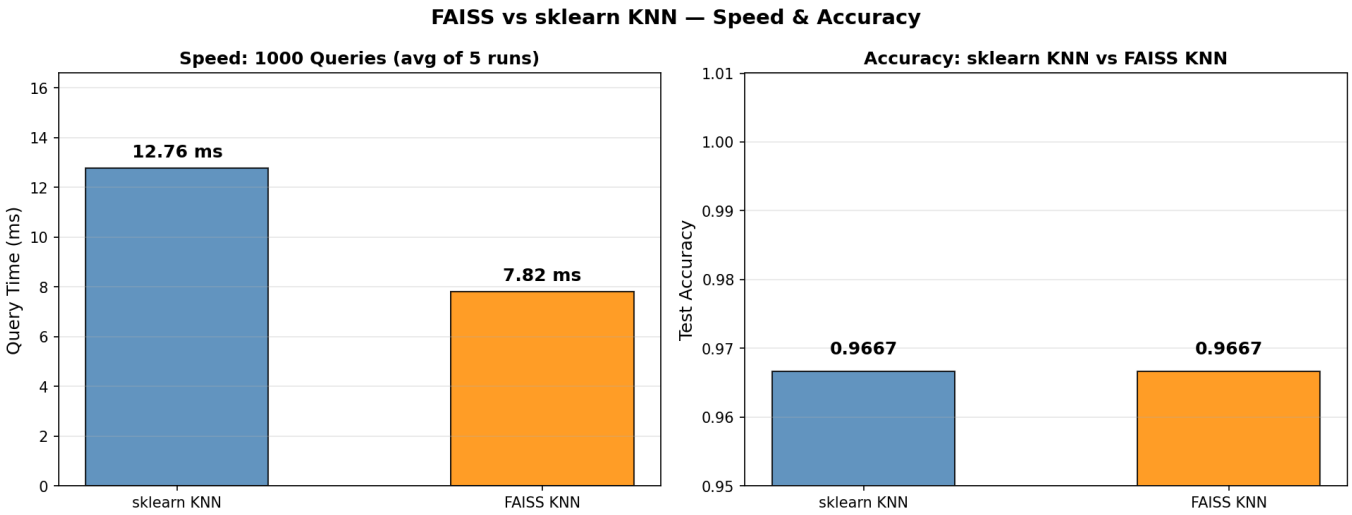

Figure 6: FAISS vs sklearn KNN — Speed & Accuracy

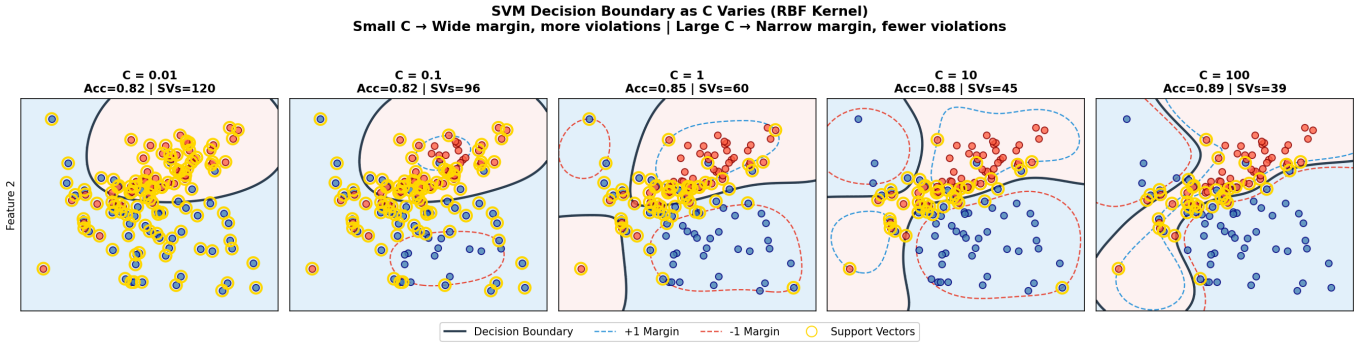

Figure 7: SVM Decision Boundary as C Varies (RBF Kernel)

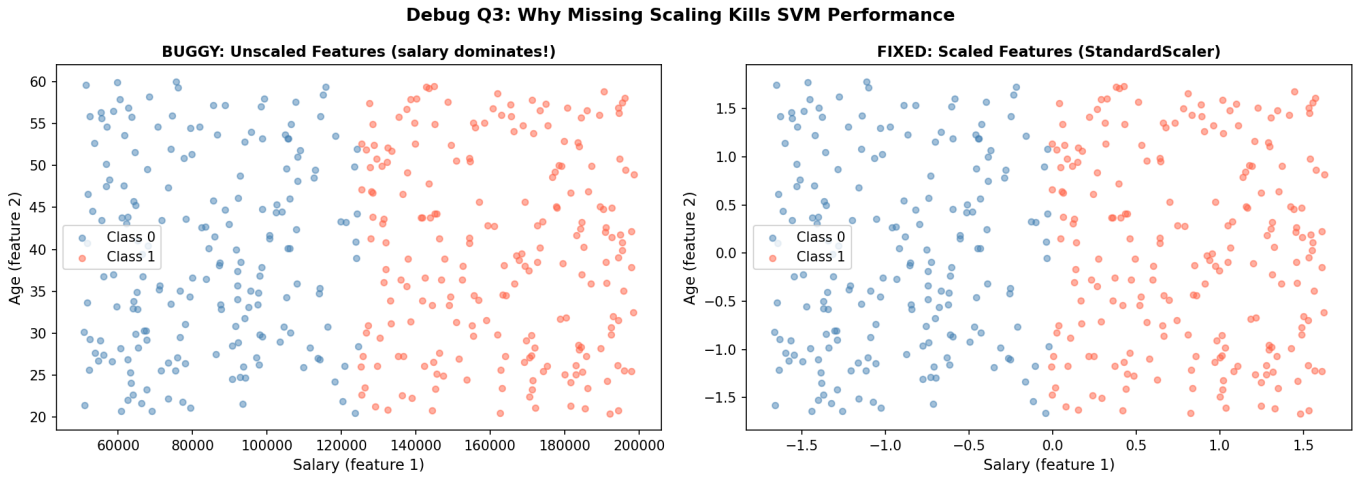

Figure 8: Debug Q3 — Effect of Missing StandardScaler